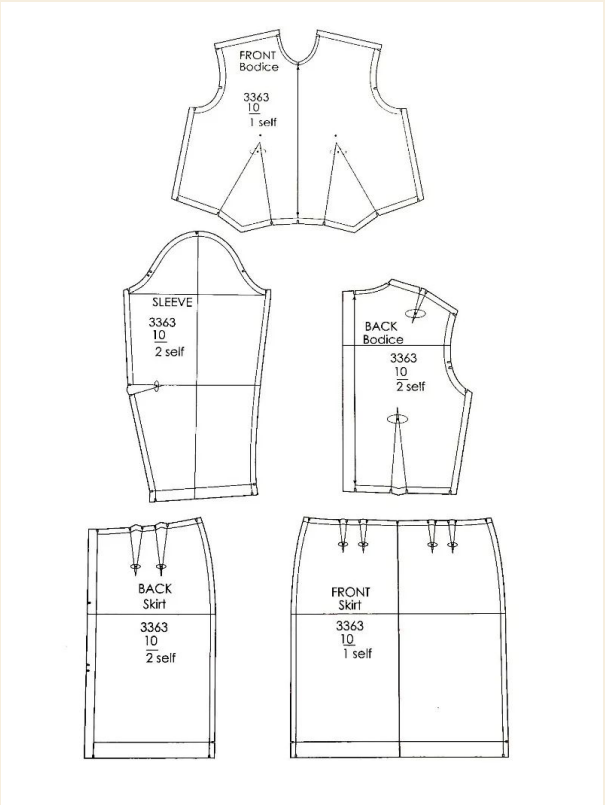
3D from 2D: flat pattern drafting and garment construction

Wendy Knox Everette - Hackers 2026

2 ways to create patterns: draping & flat pattern drafting



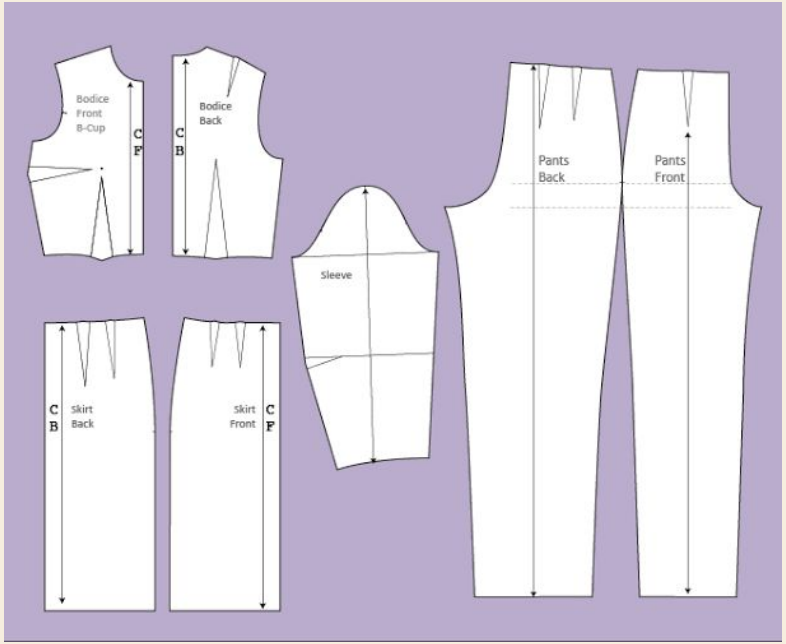


Slopers

Usually we create a **sloper** which is a very close fitting pattern that can be used as a base- we draft on top of it

https://www.sewessential.co.uk/blogs/sewing/how-to-create-and-use-a-sloper-pattern-for-perfectly-fitted-garments

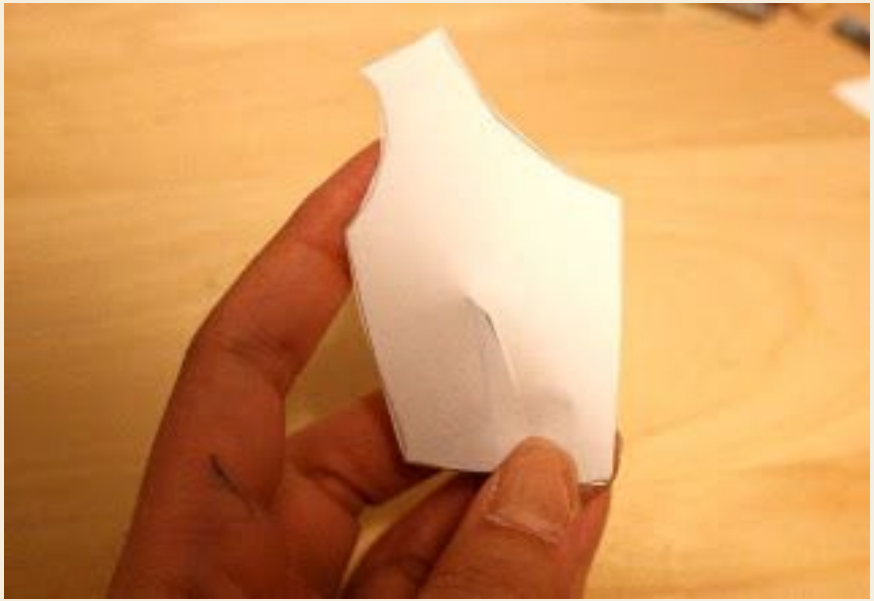
https://fashion-incubator.com/what_is_a_sloper/



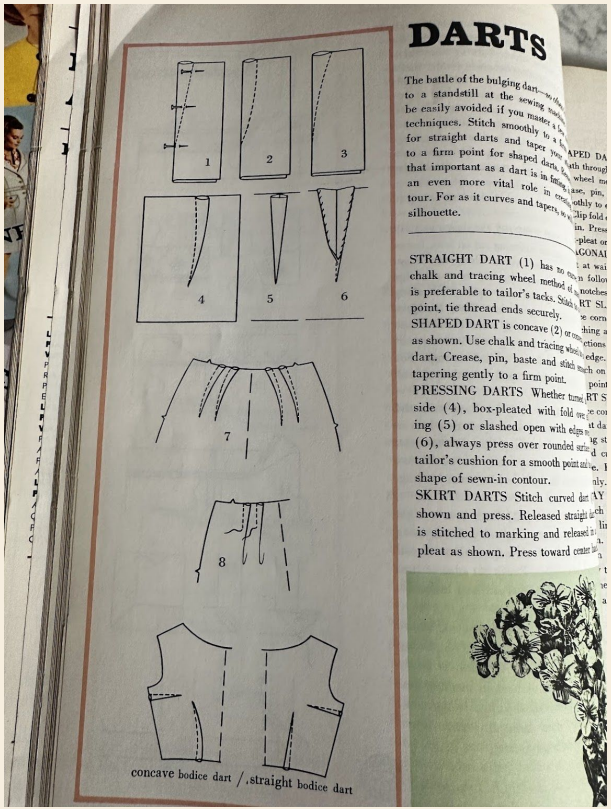
How do we make 3D shapes

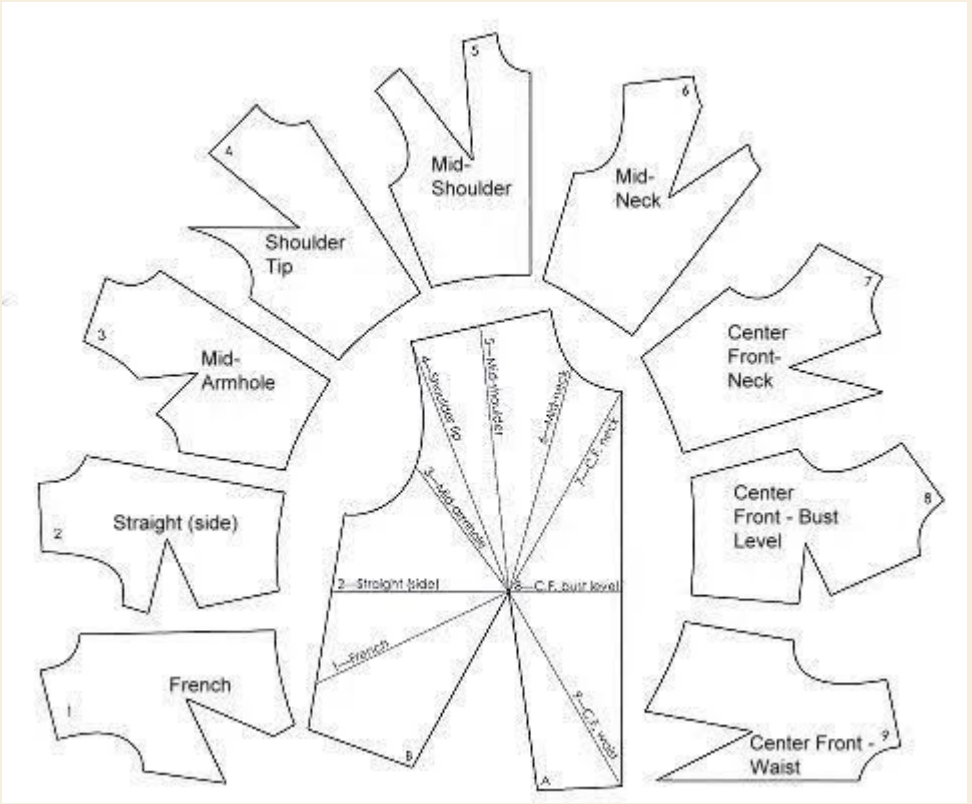
Darts



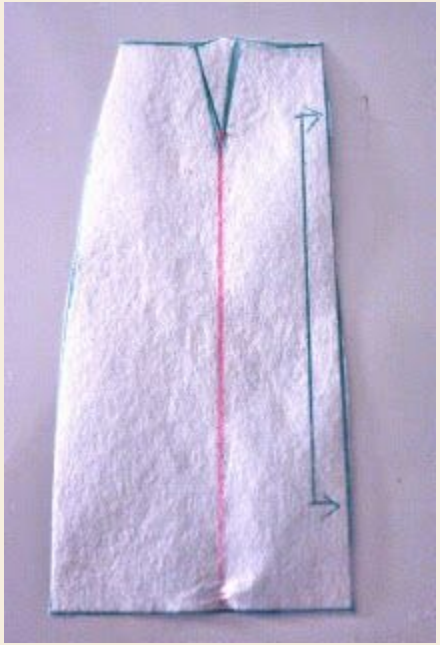


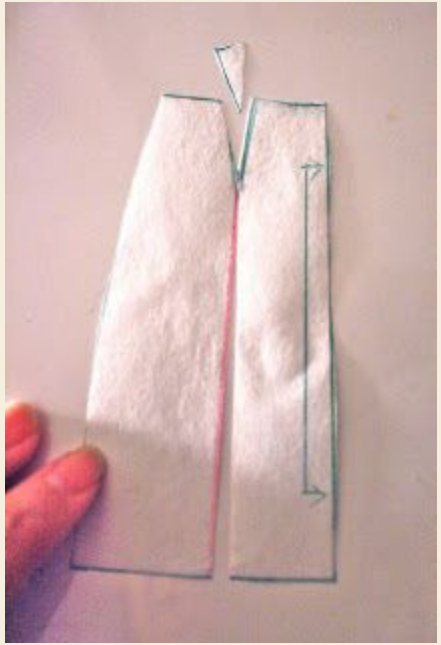
Darts can be curved or straight

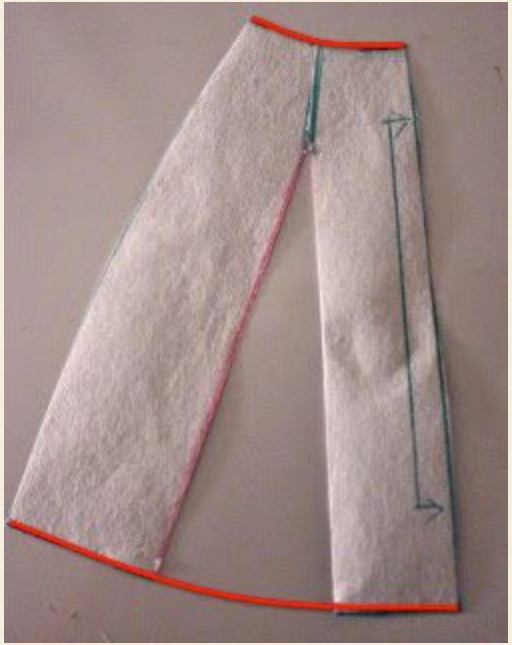




We can close darts to make new shapes

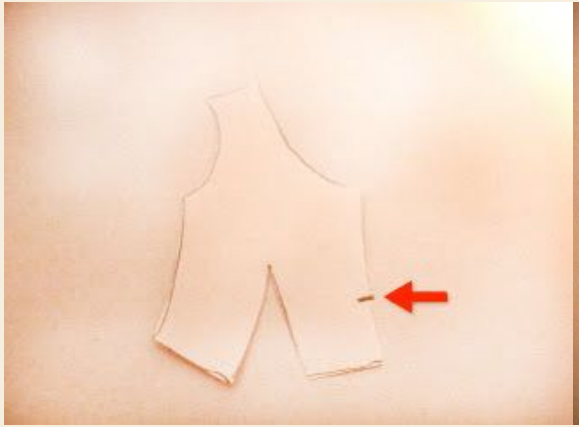


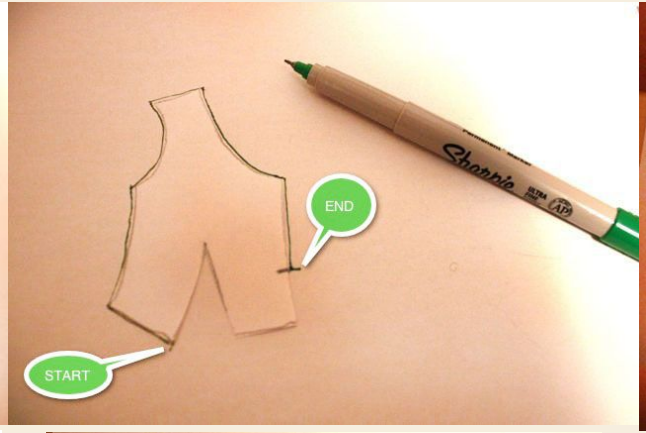


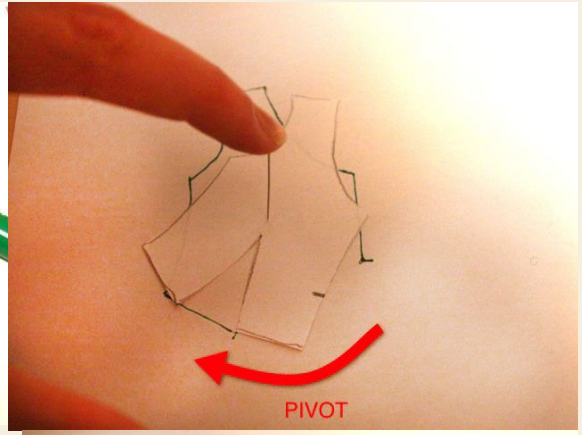


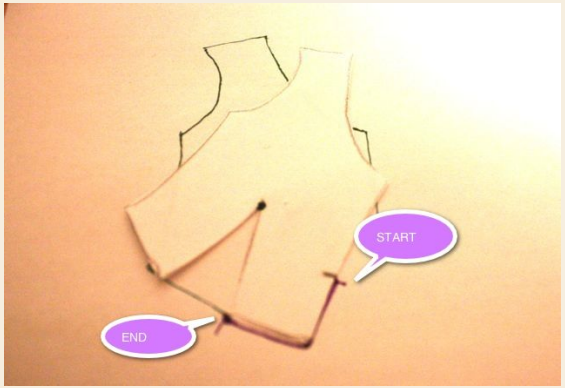
http://www.blogforbettersewing.com/2011/10/converting-skirt-dart-to-hemline-flare.html

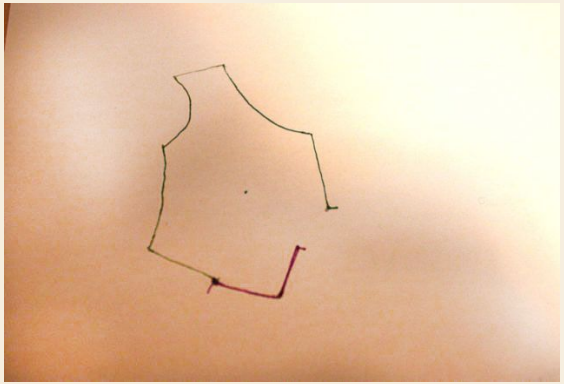
We can move darts

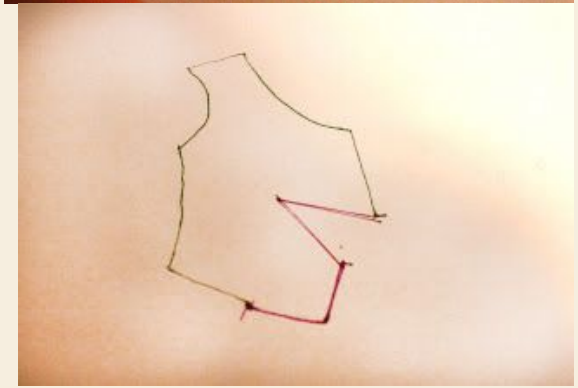




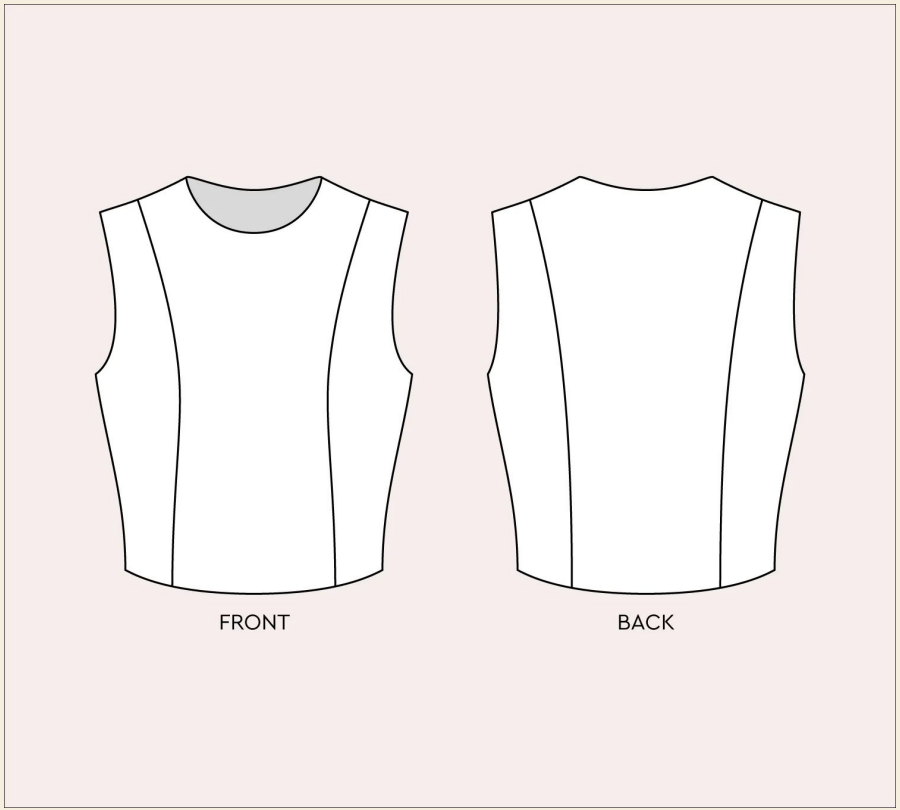


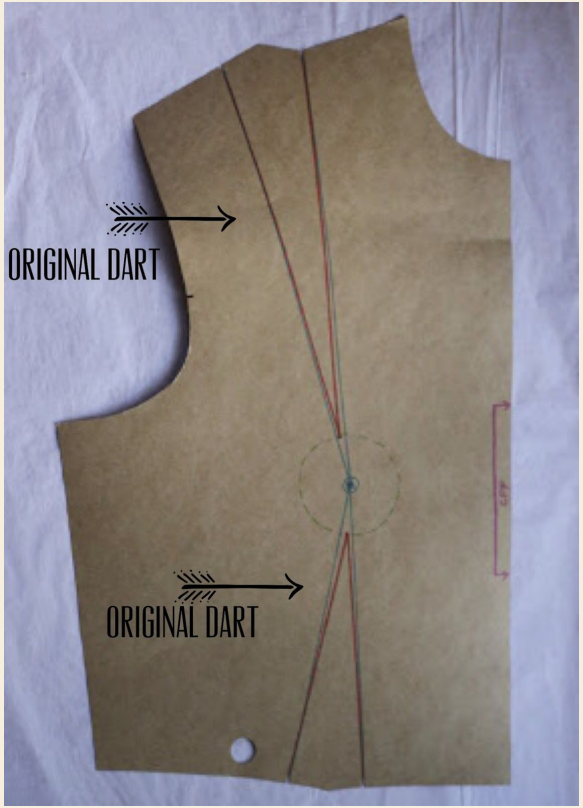


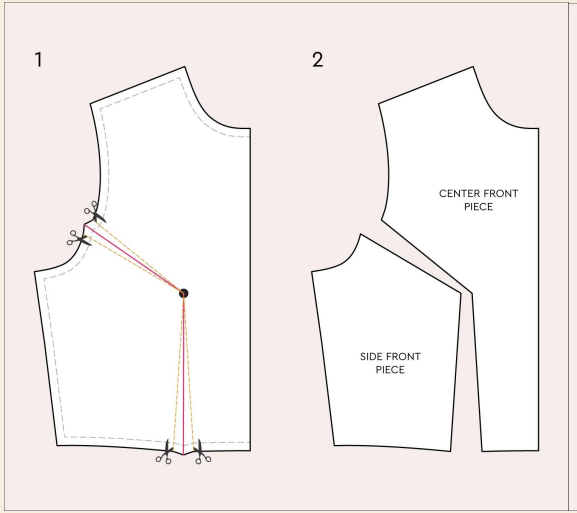


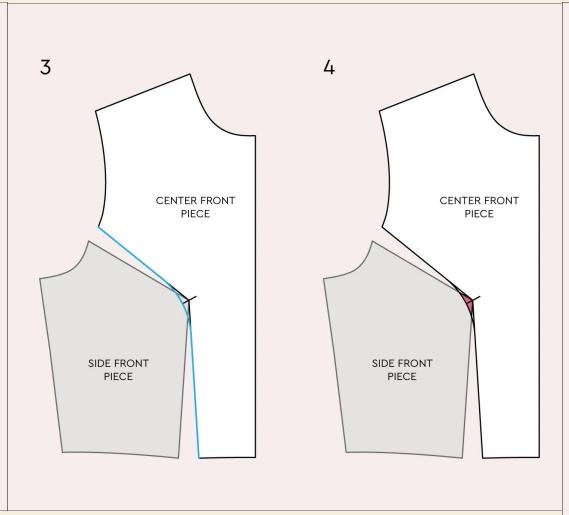


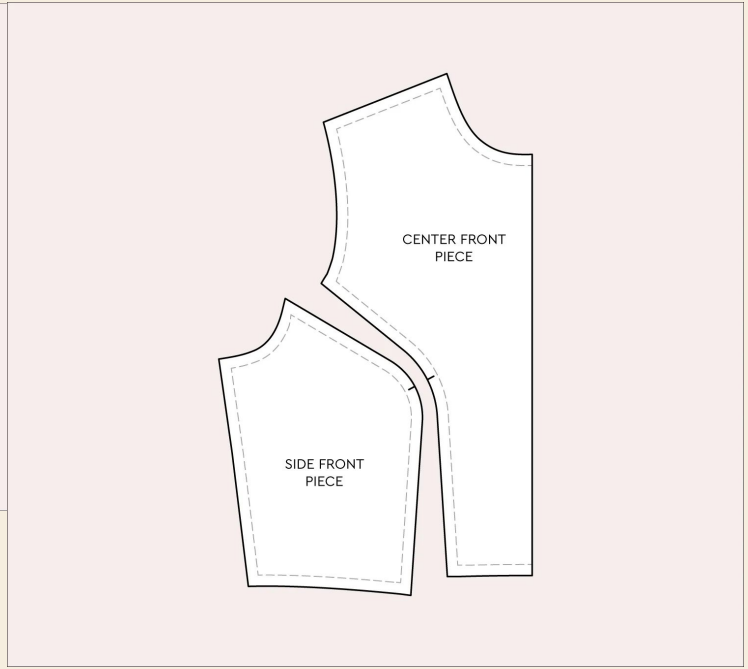
Most bodices are darts or princess seams









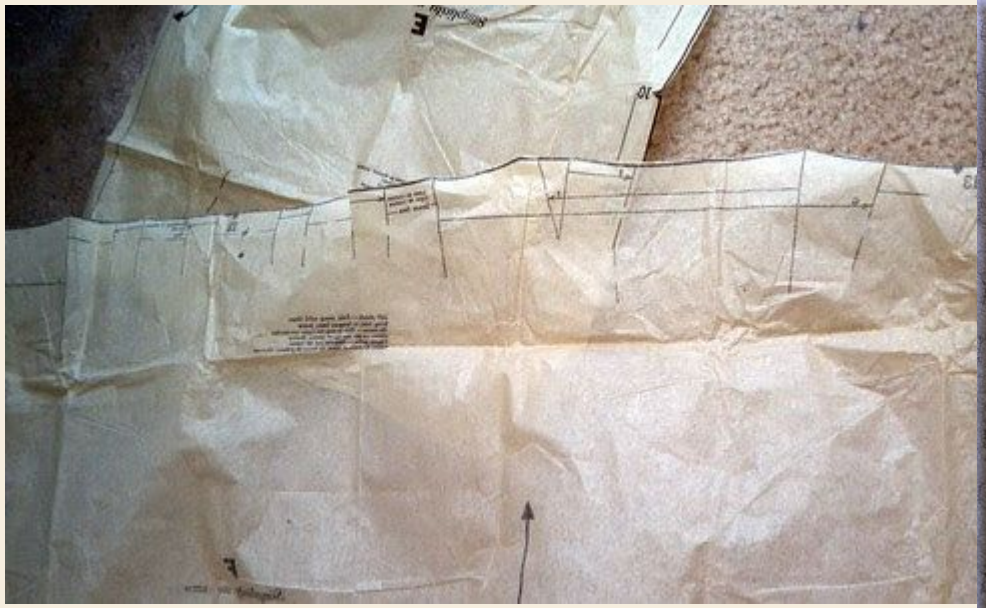


Modify darts into princess seams

https://blog.closetcorepatterns.com/bodice-fitting-4-how-to-convert-darts-to-princess-seams/

Other ways to make 3D shapes - pleats







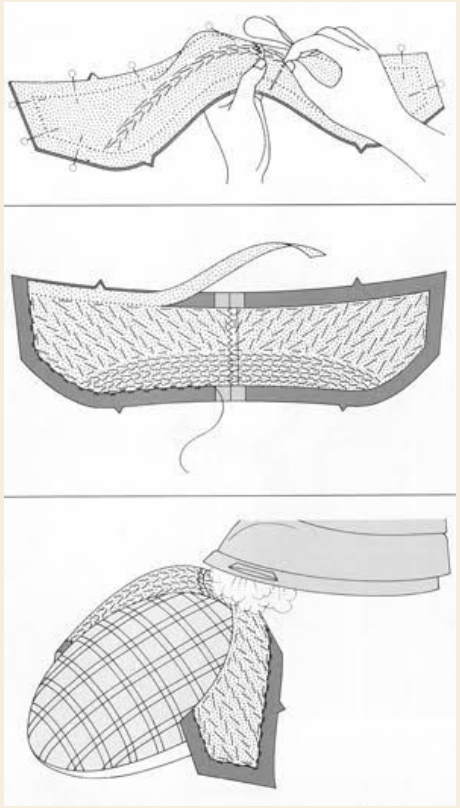
Other ways to make 3D shapes - gathers





Or manipulating the fabric





More manipulation

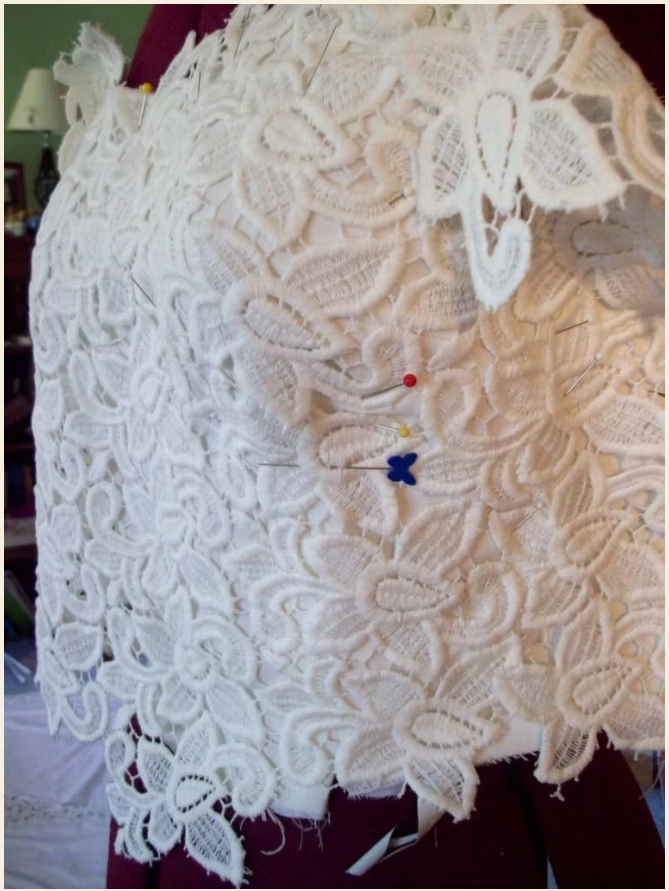






Darts & Guipure Lace

https://www.pinkhollybushdesigns.com/post/working-with-guipure-lace-forming-darts



Interfacing

Can be iron on or sewn in; you may use sheets of interfacing or tapes





https://www.sewessential.co.uk/blogs/sewing/interfacing-tapes-i-use

We combine these techniques for many garments



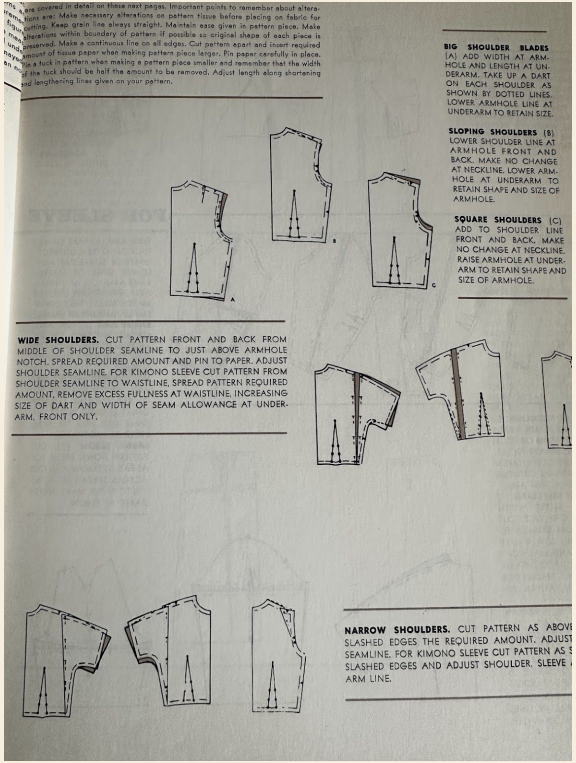


Interlining Vs Underlining

Underline: attach two fabrics and treat as one

Interlining: putting a fabric between the outer fabric and the lining (for structure, or warmth, or both)

Altering patterns for fit



are covered in detail on those next pages. Important points to remember about alterations are: Make necessary alterations on pattern tissue before placing on fabric for cutting. Keep grain line always straight. Maintain ease given in pattern piece. Make alterations within boundary of pattern if possible so original shape of each piece is preserved. Make a continuous line on all edges. Cut pattern apart and insert required amount of tissue paper when making pattern piece larger. Pin paper carefully in place. To take a tuck in pattern when making a pattern piece smaller and remember that the width of the tuck should be half the amount to be removed. Adjust length along shortening and lengthening lines given on your pattern.

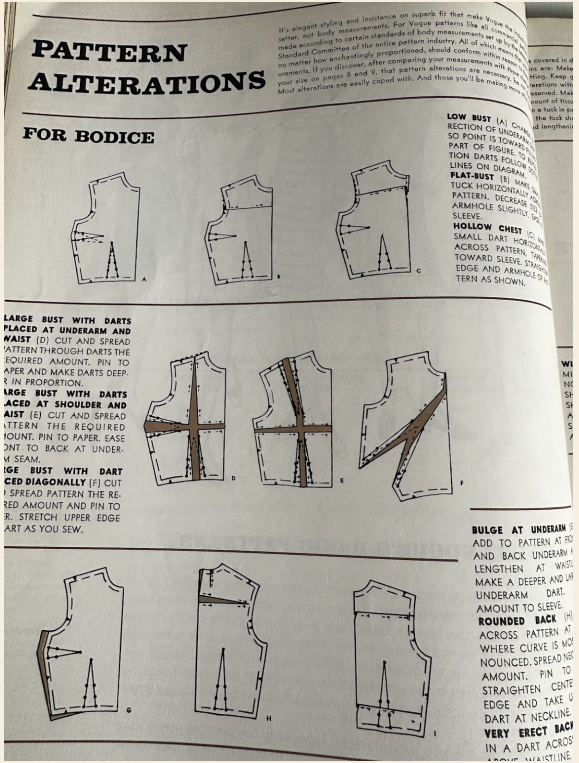
BIG SHOULDER BLADES (A) ADD WIDTH AT ARMHOLE AND LENGTH AT UNDERARM. TAKE UP A DART ON EACH SHOULDER AS SHOWN BY DOTTED LINES. LOWER ARMHOLE LINE AT UNDERARM TO RETAIN SIZE.

SLOPING SHOULDERS (B) LOWER SHOULDER LINE AT ARMHOLE FRONT AND BACK. MAKE NO CHANGE AT NECKLINE. LOWER ARMHOLE AT UNDERARM TO RETAIN SHAPE AND SIZE OF ARMHOLE.

SQUARE SHOULDERS (C) ADD TO SHOULDER LINE FRONT AND BACK. MAKE NO CHANGE AT NECKLINE. RAISE ARMHOLE AT UNDERARM TO RETAIN SHAPE AND SIZE OF ARMHOLE.

WIDE SHOULDERS. CUT PATTERN FRONT AND BACK FROM MIDDLE OF SHOULDER SEAMLINE TO JUST ABOVE ARMHOLE NOTCH. SPREAD REQUIRED AMOUNT AND PIN TO PAPER. ADJUST SHOULDER SEAMLINE. FOR KIMONO SLEEVE CUT PATTERN FROM SHOULDER SEAMLINE TO WAISTLINE. SPREAD PATTERN REQUIRED AMOUNT. REMOVE EXCESS FULLNESS AT WAISTLINE, INCREASING SIZE OF DART AND WIDTH OF SEAM ALLOWANCE AT UNDERARM. FRONT ONLY.

NARROW SHOULDERS. CUT PATTERN AS ABOVE SLASHED EDGES THE REQUIRED AMOUNT. ADJUST SEAMLINE. FOR KIMONO SLEEVE CUT PATTERN AS SLASHED EDGES AND ADJUST SHOULDER. SLEEVE ARM LINE.



PATTERN ALTERATIONS

FOR BODICE

It's elegant styling and insistence on superb fit that make Vogue patterns the perfect... not body measurements... made according to certain standards of body measurements... Standard Committee of the entire pattern industry. All of which means... no matter how enchantingly proportioned, should conform within reason... garments. If you discover, after comparing your measurements with those... your size on pages 8 and 9, that pattern alterations are necessary... Most alterations are easily coped with. And those you'll be making...

LOW BUST (A) ... RECTION OF UNDER... SO POINT IS TOWARD... PART OF FIGURE. ... TION DARTS FOLLOW... LINES ON DIAGRAM.

FLAT-BUST (B) MAKE ... TUCK HORIZONTALLY ... PATTERN. DECREASE ... ARMHOLE SLIGHTLY ... SLEEVE.

HOLLOW CHEST ... SMALL DART HORIZ... ACROSS PATTERN, TAPERING TOWARD SLEEVE. STRAIGHTEN EDGE AND ARMHOLE OF PATTERN AS SHOWN.

LARGE BUST WITH DARTS PLACED AT UNDERARM AND WAIST (D) CUT AND SPREAD PATTERN THROUGH DARTS THE REQUIRED AMOUNT. PIN TO PAPER AND MAKE DARTS DEEPER IN PROPORTION.

LARGE BUST WITH DARTS PLACED AT SHOULDER AND WAIST (E) CUT AND SPREAD PATTERN THE REQUIRED AMOUNT. PIN TO PAPER. EASE FRONT TO BACK AT UNDERARM SEAM.

LARGE BUST WITH DART PLACED DIAGONALLY (F) CUT AND SPREAD PATTERN THE REQUIRED AMOUNT AND PIN TO PAPER. STRETCH UPPER EDGE OF DART AS YOU SEW.

BULGE AT UNDERARM ... ADD TO PATTERN AT FRONT AND BACK UNDERARM ... LENGTHEN ... AT WAIST... MAKE A DEEPER AND LARGER UNDERARM DART. ... AMOUNT TO SLEEVE.

ROUNDED BACK (H) ... ACROSS PATTERN AT ... WHERE CURVE IS MOST PRONOUNCED. SPREAD REQUIRED AMOUNT. PIN TO PAPER. STRAIGHTEN CENTER EDGE AND TAKE UP DART AT NECKLINE.

VERY ERECT BACK ... IN A DART ACROSS ... ABOVE WAISTLINE.


Machado's Plane Talk
BOUT FACE



